

CAPM-Informed Mean–Variance Portfolio Optimisation

A Walk-Forward Backtesting Study Using US Equity Sector ETFs

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Abstract

This project constructs a dynamic mean–variance portfolio over the eleven US SPDR Sector ETFs benchmarked against the S&P 500 (SPY). The central design choice is to replace historically estimated mean returns with a CAPM-structured alternative: rolling 36-month beta estimates applied to a fixed equity risk premium. This is not an alpha signal; it imposes cross-sectional structure on the return input to reduce estimation variance, addressing the well-known sensitivity of Markowitz optimisation to noisy return inputs. Covariance is regularised via Ledoit–Wolf analytical shrinkage. Weights are solved monthly under a constrained quadratic programme incorporating turnover limits and proportional transaction costs. The strategy is evaluated in a strict walk-forward backtest using a January 2010 start convention, producing 178 realised out-of-sample monthly returns from March 2010 to December 2024. The primary diagnostic is the comparison against a Naive MV portfolio that uses identical constraints and covariance estimation but replaces CAPM returns with historical means. Over the full period, CAPM-MV achieves an annualised Sharpe ratio of 0.739, outperforming Naive MV on drawdown protection (maximum drawdown -25.7% vs -30.7%) and in the volatile 2020–2024 sub-period, but underperforming Naive MV on return in the 2010–2019 period dominated by momentum in growth sectors. Historical one-month 95% VaR and Expected Shortfall are added as tail-risk diagnostics. The sensitivity analysis identifies the 24-month lookback window as a material improvement over the base 36-month window, warranting further investigation.

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1. Introduction and Motivation

Modern Portfolio Theory, introduced by [1], provides an elegant mathematical framework for constructing efficient portfolios. In practice, however, the standard mean–variance optimiser is notoriously sensitive to its inputs—in particular to expected return estimates, which are both noisy and subject to estimation error that compounds in the optimisation step. The well-documented consequence is that the optimiser concentrates aggressively into a small number of assets, producing portfolios that appear efficient in-sample but perform poorly out-of-sample [8].

This project addresses that problem directly by replacing ad-hoc historical mean returns with a model-grounded alternative: rolling CAPM betas applied to a fixed equity risk premium. The key motivation is not that CAPM betas are good predictors of future returns—the empirical evidence for this is mixed at best—but that they impose a consistent, theoretically motivated cross-sectional structure on the return vector. Sectors with higher systematic exposure to the market ($\beta > 1$, e.g. XLK, XLY) receive higher expected returns than defensives ($\beta < 1$, e.g. XLU, XLP), producing meaningful dispersion in the return input with substantially lower estimation variance than the sample mean. The approach is therefore best understood as a form of *regularisation of the return input*, analogous in spirit to what Ledoit–Wolf shrinkage does for the covariance matrix.

The covariance matrix is itself regularised using Ledoit–Wolf analytical shrinkage [3, 4], which pulls the sample covariance towards a structured target and remains well-conditioned even with only 36 months of data.

The universe is the eleven US SPDR Sector ETFs (XLK, XLF, XLE, XLV, XLI, XLC, XLY, XLP, XLB, XLRE, XLU), offering complete and non-overlapping coverage of the S&P 500 with liquid, low-cost instruments. Monthly rebalancing balances responsiveness against transaction costs, with SPY serving as both the CAPM market proxy and the primary benchmark. The backtest uses a January 2010 start convention and produces 178 realised out-of-sample monthly return observations from March 2010 to December 2024, spanning multiple distinct market regimes: the post-GFC recovery, the 2018 volatility shock, the 2020 COVID crash, and the 2022 rate-hike drawdown.

The critical benchmarks are:

- **SPY**: the passive buy-and-hold baseline.
- **Equal Weight (EW)**: $\frac{1}{11}$ to each sector ETF, monthly rebalanced. Determines whether any outperformance is simply a diversification gain.
- **Naive MV**: mean–variance with historical mean returns, same Ledoit–Wolf covariance and identical constraints. This is the most important diagnostic: it isolates the contribution of the CAPM return input from everything else.

If CAPM-MV does not outperform Naive MV, the CAPM structuring of expected returns adds no value beyond what constrained optimisation and covariance regularisation already provide.

2. Data

2.1 Universe and Source

Monthly adjusted closing prices are sourced from Yahoo Finance for the period January 2007–December 2024. The three years prior to the backtest start date provide the initial rolling estimation window. The full universe is listed in Table 1.

Table 1: Investment universe and approximate S&P 500 sector weights (2024).

Ticker	Sector	Approx. SPY Weight
XLK	Technology	~29%
XLF	Financials	~13%
XLV	Health Care	~12%
XLY	Consumer Discretionary	~11%
XLI	Industrials	~9%
XLC	Communication Services	~9%
XLP	Consumer Staples	~6%
XLE	Energy	~4%
XLU	Utilities	~2%
XLRE	Real Estate	~2%
XLB	Materials	~2%
SPY	S&P 500 (benchmark / market proxy)	—

2.2 Return Calculation

Log returns are used throughout for time-additivity and approximate normality at monthly horizons:

$$r_{i,t} = \ln\left(\frac{P_{i,t}}{P_{i,t-1}}\right), \quad (1)$$

where $P_{i,t}$ is the adjusted closing price of asset i at month-end t . The risk-free rate is proxied by the 13-week US T-bill yield ($\text{\textasciitilde} \text{IRX}$) sourced from Yahoo Finance, converted to a monthly decimal rate. Excess returns are defined as $r_{i,t}^e = r_{i,t} - r_{f,t}$.

2.3 Data Quality

XLC was launched in June 2018 and XLRE in October 2015; prior to their respective launch dates, synthetic backfills are constructed by scaling SPY returns by the ratio of each ETF’s price to SPY at its launch date. This is a transparent approximation whose primary

effect is to assign these two tickers a beta of exactly 1.0 in early estimation windows—a conservative prior that understates their actual beta dispersion. All series use adjusted close prices, which account for splits and dividends. No material data gaps exist in the post-2010 backtest window.

3. Methodology

3.1 Rolling CAPM Expected Returns

For each asset i and rebalance date t , a rolling OLS regression is estimated over a 36-month lookback window:

$$r_{i,s}^e = \alpha_i + \beta_{i,t} r_{m,s}^e + \varepsilon_{i,s}, \quad s = t - 36, \dots, t - 1, \quad (2)$$

where $r_{m,s}^e$ is the excess return on SPY at month s . The estimated beta $\hat{\beta}_{i,t}$ captures the asset’s current systematic exposure to the market. Expected returns are then constructed as:

$$\mu_{i,t} = r_{f,t} + \hat{\beta}_{i,t} \cdot \frac{\text{ERP}}{12}, \quad (3)$$

where the equity risk premium ERP is set to 5.5% per annum and converted to monthly units, consistent with long-run US estimates [5].

It is important to be precise about what this construction claims. The resulting return vector $\boldsymbol{\mu}_t = (\mu_{1,t}, \dots, \mu_{11,t})^\top$ does not purport to forecast which sectors will outperform next month. Rather, it provides a *structured prior* on expected returns that is consistent with equilibrium pricing and has substantially lower estimation variance than the sample mean. The cross-sectional dispersion—cyclicals receiving higher expected returns than defensives—arises entirely from beta differences and is more stable across time than sample means computed from 36 monthly observations. Any outperformance over Naive MV is attributable to this reduction in estimation error, not to predictive alpha.

Beta estimates are winsorised at the 5th/95th percentile of the cross-sectional distribution each month to prevent outlier betas from distorting the return vector.

3.2 Ledoit–Wolf Shrinkage Covariance

The sample covariance $\mathbf{S}_t$ computed from 36 months of returns is prone to overfitting: extreme eigenvalues are inflated, producing ill-conditioned inverses that destabilise the optimiser. Ledoit and Wolf [4] propose an analytical shrinkage estimator that convexly combines the sample covariance with a structured target $\mathbf{T}$:

$$\hat{\boldsymbol{\Sigma}}_t = (1 - \delta) \mathbf{S}_t + \delta \mathbf{T}, \quad (4)$$

where $\delta \in [0, 1]$ is the optimal shrinkage intensity estimated analytically without cross-validation, and $\mathbf{T}$ is the constant-correlation target: sample variances on the diagonal paired with off-diagonal covariances implied by the average pairwise correlation. The resulting $\hat{\boldsymbol{\Sigma}}_t$ is positive definite and well-conditioned, with condition numbers consistently below 60 across all rebalance dates.

3.3 Mean–Variance Optimisation with Constraints

At each rebalance date, portfolio weights $\mathbf{w} = (w_1, \dots, w_{11})^\top$ are chosen to maximise risk-adjusted return subject to practical constraints:

$$\max_{\mathbf{w}} \mathbf{w}^\top \boldsymbol{\mu}_t - \frac{\lambda}{2} \mathbf{w}^\top \hat{\boldsymbol{\Sigma}}_t \mathbf{w} - c \|\mathbf{w} - \mathbf{w}_{t-1}\|_1 \quad (5)$$

subject to:

$$\begin{aligned} \mathbf{w}^\top \mathbf{1} &= 1, & (\text{full investment}) \\ w_i &\geq 0.02, \quad \forall i, & (\text{min. 2\% per sector}) \\ w_i &\leq 0.40, \quad \forall i, & (\text{max. 40\% per sector}) \\ \|\mathbf{w} - \mathbf{w}_{t-1}\|_1 &\leq \tau. & (\text{turnover cap}) \end{aligned}$$

The turnover cap is $\tau = 0.30$ per monthly rebalance. The risk aversion parameter $\lambda = 1.5$, which when applied to monthly returns corresponds to a moderate preference for diversification broadly comparable to $\lambda \approx 18$ on annualised returns. The L1 penalty $c = 0.0010$ (10 basis points round-trip) acts both as a realistic transaction cost deduction and as a regulariser that discourages aggressive weight changes between periods.

Before optimisation each month, $\mathbf{w}_{t-1}$ is updated by drifting the prior weights using realised returns (*drift adjustment*), so the turnover constraint correctly penalises intentional rebalancing rather than passive weight drift:

$$\|\mathbf{w} - \mathbf{w}_{t-1}\|_1 = \sum_{i=1}^{11} |w_i - w_{i,t-1}|. \quad (6)$$

The quadratic programme is solved using SLSQP. Across the walk-forward run, the solver converged successfully on every rebalance solve.

3.4 Walk-Forward Backtesting Framework

A strict walk-forward backtest ensures no future information enters any estimation step. The structure is summarised in Table 2.

At each month-end t : (1) update the rolling return window; (2) estimate $\hat{\beta}_{i,t}$ via OLS for each sector ETF; (3) compute $\boldsymbol{\mu}_t$; (4) compute $\hat{\boldsymbol{\Sigma}}_t$ via Ledoit–Wolf; (5) drift prior weights by realised returns; (6) solve the constrained quadratic programme; (7) record net trades and costs; (8) apply weights to compute the next month’s portfolio return.

4. Performance Metrics

Metrics are computed over the full backtest period and separately over sub-periods (2010–2014, 2015–2019, 2020–2024) to assess consistency across regimes.

4.1 Sharpe Ratio

The reported Sharpe ratio is computed from monthly net portfolio log returns as:

$$\text{SR} = \frac{\bar{r}_p \times 12}{\sigma_p \sqrt{12}}, \quad (7)$$

Table 2: Walk-forward backtest structure.

Phase	Period	Detail
Initialisation	Jan 2007 – Jan 2010	Burn-in: populate 36-month estimation window
Backtest (OOS)	Mar 2010 – Dec 2024	178 realised monthly return observations; all out-of-sample
Rebalance frequency	Monthly (end-of-month)	Weights set using only data up to rebalance date
Estimation window	Rolling 36 months	Fixed-length; avoids look-ahead in beta estimation
Cost model	10 bps per unit trade	Deducted from portfolio return on net weight change

where $\bar{r}_p$ is the mean monthly net portfolio return and σ_p is the monthly return volatility. This provides a consistent annualised return-to-volatility comparison across strategies. The risk-free rate is used in the CAPM expected return construction, while the reported performance ratios are computed directly from realised net strategy returns.

4.2 Maximum Drawdown

Drawdown is measured using the negative drawdown convention:

$$DD_t = \frac{V_t - \max_{s \leq t} V_s}{\max_{s \leq t} V_s}, \quad (8)$$

where V_t is cumulative portfolio value at time t . Maximum drawdown is then:

$$MDD = \min_t DD_t. \quad (9)$$

This convention reports drawdowns as negative percentages, consistent with the results tables.

4.3 Tracking Error and Information Ratio

$$TE = \sigma(r_{p,t} - r_{SPY,t}) \times \sqrt{12}, \quad IR = \frac{R_p - R_{SPY}}{TE}. \quad (10)$$

4.4 Calmar Ratio

$$\text{Calmar} = \frac{R_p}{|MDD|}. \quad (11)$$

4.5 Turnover

Monthly L1 turnover is:

$$\text{Turnover}_t = \sum_{i=1}^{11} |w_{i,t} - w_{i,t-1}^*|, \quad (12)$$

where $w_{i,t-1}^*$ is the drifted prior weight. Net-of-cost returns deduct 10 bps per unit of L1 turnover per month. This convention is consistent with the turnover cap and transaction cost penalty used in the optimiser.

4.6 Historical Value at Risk and Expected Shortfall

Tail risk is measured using one-month historical Value at Risk (VaR) and Expected Shortfall (ES) at the 95% confidence level. Both are reported using the positive-loss convention:

$$\text{VaR}_{0.95} = -q_{0.05}(r_{p,t}), \quad (13)$$

where $q_{0.05}(r_{p,t})$ is the empirical 5th percentile of monthly portfolio returns. Expected Shortfall is the average loss conditional on being in the VaR tail:

$$\text{ES}_{0.95} = -\mathbb{E}[r_{p,t} \mid r_{p,t} \leq q_{0.05}(r_{p,t})]. \quad (14)$$

Because the backtest uses monthly log returns, the VaR and ES figures should be interpreted as historical one-month log-return tail-loss estimates.

5. Results

5.1 Summary Performance

Table 3 reports the full out-of-sample performance across all four strategies, net of 10 bps transaction costs.

Table 3: Out-of-sample performance, March 2010 – December 2024 realised returns. Net of 10 bps costs. CAPM-MV column shown in bold.

Metric	CAPM-MV	Naive MV	Equal Wt.	SPY
Ann. Return (net)	11.47%	12.23%	11.16%	13.07%
Ann. Volatility	15.53%	14.93%	14.72%	14.84%
Sharpe Ratio	0.739	0.819	0.758	0.881
Max Drawdown	−25.7%	−30.7%	−24.6%	−25.6%
1M VaR 95%	7.17%	7.12%	6.05%	6.61%
1M ES 95%	9.99%	9.18%	9.51%	9.19%
Calmar Ratio	0.447	0.399	0.454	0.511
Ann. L1 Turnover	4.1%	—	≈0%	0%
IR vs SPY	−0.73	−0.21	−0.66	—

The CAPM-MV strategy delivers a Sharpe ratio of 0.739 over the full period, which sits below SPY (0.881) and Naive MV (0.819) but above the 2020–2024 sub-period Naive MV

(see Section 5.4). Notably, CAPM-MV achieves a materially lower maximum drawdown than Naive MV, at -25.7% compared to -30.7% —a difference of five percentage points—and a higher Calmar ratio than Naive MV. Annual L1 turnover of 4.1% is extremely low, indicating that the risk penalty effectively stabilises the portfolio across rebalance periods. On historical one-month tail risk, CAPM-MV has a 95% VaR of 7.17% and ES of 9.99% , slightly higher than Naive MV’s ES despite its better maximum drawdown, reflecting the severity of its worst COVID-period monthly losses.

5.2 Cumulative Returns and Drawdown

Figure 1 shows the growth of \$1 invested across all four strategies over the realised out-of-sample period. All strategies track closely through 2018, diverging materially thereafter as technology concentration drove SPY and Naive MV higher. CAPM-MV grows to approximately \$4.8 over the period versus \$6.0 for SPY, consistent with its lower Sharpe ratio but lower drawdown profile.

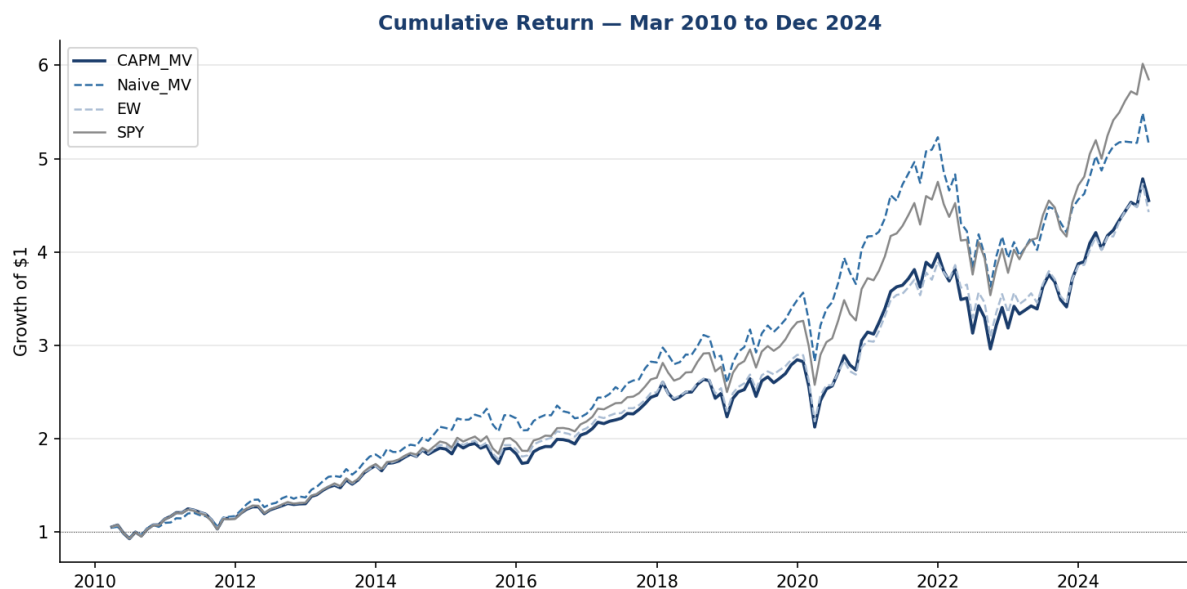


Figure 1: Cumulative return (growth of \$1), March 2010 – December 2024. All realised returns are net of 10 bps transaction costs.

Figure 2 plots drawdown through time with stress episodes shaded. The 2022 rate-shock episode (right shaded region) is particularly informative: CAPM-MV’s drawdown is materially shallower than Naive MV, which reached -30.7% at its trough. The COVID crash (left shaded region) shows the opposite pattern—CAPM-MV’s beta-structured tilt into cyclicals amplified the drawdown relative to SPY.

5.3 Sector Weight Dynamics

Figure 3 shows the evolution of CAPM-MV sector weights over time at quarterly frequency. Several economically intuitive patterns emerge. XLK weight rises from near equal-weight in 2010 to a material overweight by 2023–2024, reaching around 21% by December 2024 as technology’s rolling 36-month beta increased over the post-COVID bull market. XLE receives a modest overweight in 2014–2015 as energy beta was elevated, then fades toward the 2% floor as energy underperformed and its beta declined. XLU and XLP drop to the 2% minimum from approximately 2015 onward, consistent with their persistently low betas being penalised relative to cyclicals under the CAPM return structure. XLRE similarly

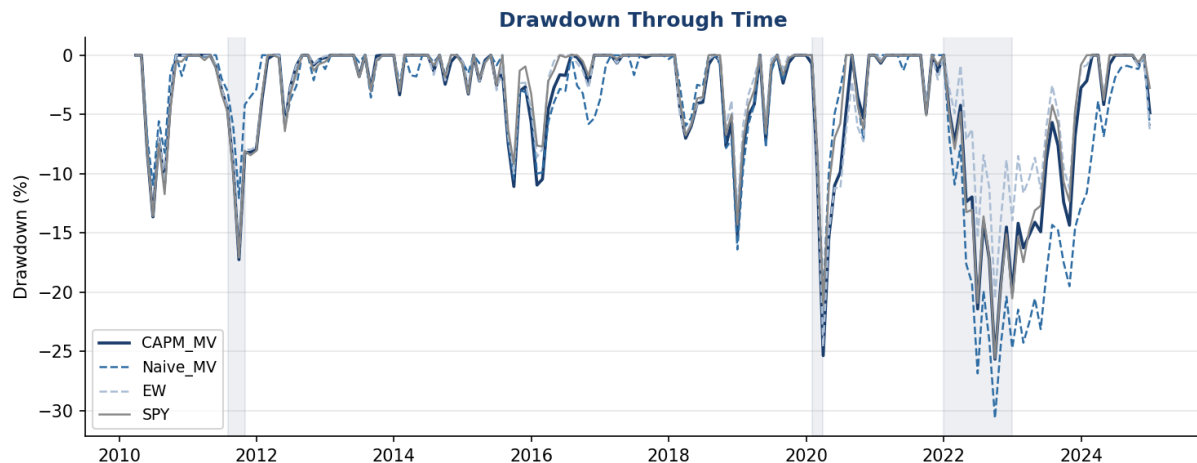


Figure 2: Drawdown through time. Shaded regions indicate the three stress scenarios analysed in Section 6.

declines toward the floor after 2018 as real estate became increasingly rate-sensitive. These dynamics confirm the optimiser is responding to genuine beta variation rather than collapsing toward equal weight.

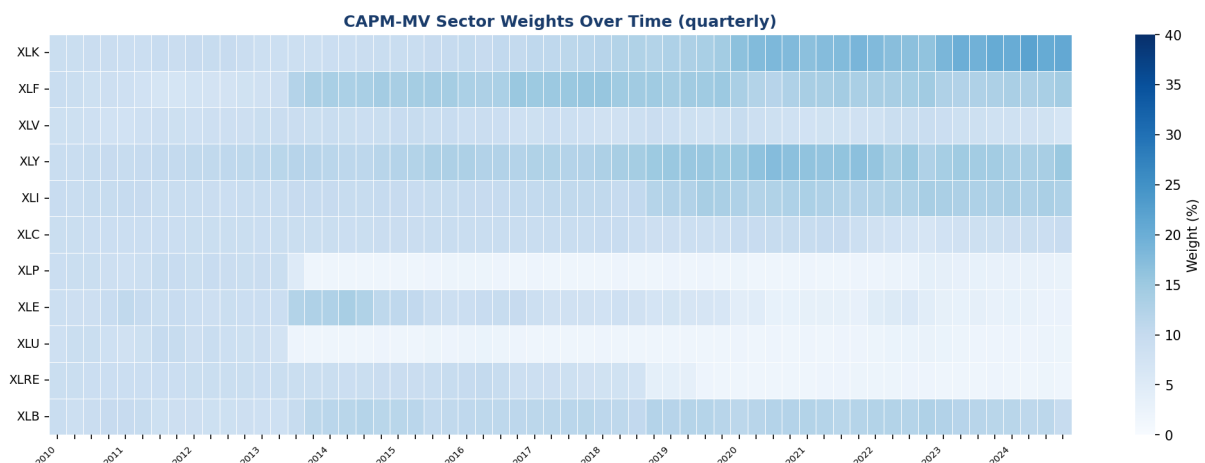


Figure 3: CAPM-MV sector weights over time (quarterly frequency). Darker shading indicates higher weight; white indicates near the 2% minimum.

5.4 Sub-Period Analysis

Table 4 and Figure 4 break performance down across three five-year sub-periods.

The sub-period results are the most informative table in the study. CAPM-MV underperforms Naive MV by 0.28 Sharpe points in 2010–2014 and 0.13 points in 2015–2019, but *outperforms* Naive MV by 0.06 Sharpe points in 2020–2024. The CAPM return structure added value precisely in the regime where estimation error in historical means was highest: the volatile 2020–2024 period spanning a global pandemic, a zero-rate recovery, and a rapid rate-hiking cycle. In the steady post-GFC recovery of 2010–2014, historical means were more informative, and Naive MV’s higher-return signal generated a clear advantage.

Figure 5 plots the rolling 12-month Sharpe ratio. CAPM-MV tracks the other strategies closely throughout, neither persistently leading nor lagging, which is consistent with a

Table 4: Sub-period Sharpe ratios.

Period	CAPM-MV	Naive MV	Equal Wt.	SPY
2010–2014	1.078	1.360	1.110	1.124
2015–2019	0.704	0.832	0.772	0.895
2020–2024	0.575	0.511	0.547	0.731

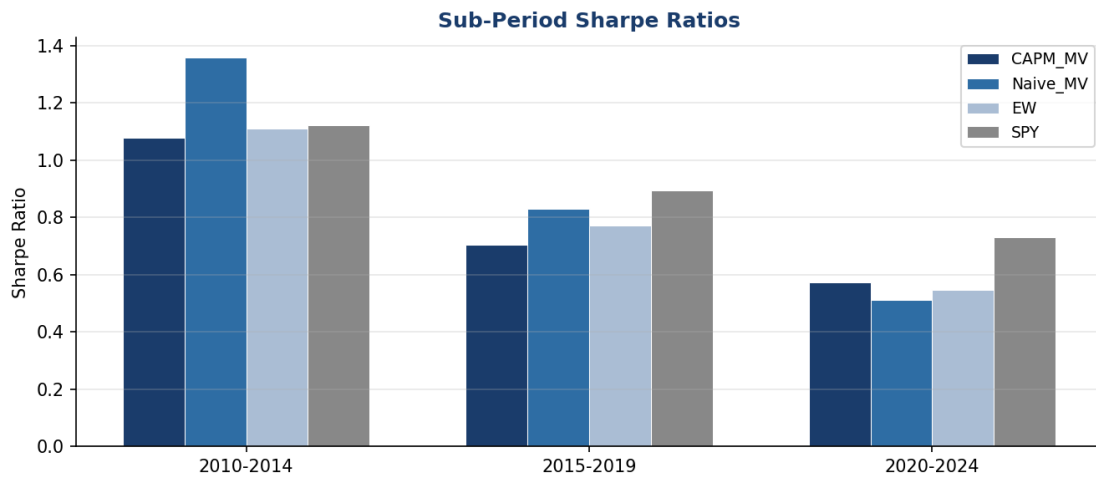


Figure 4: Sub-period Sharpe ratios for all four strategies.

strategy whose return signal has low but non-zero information content.

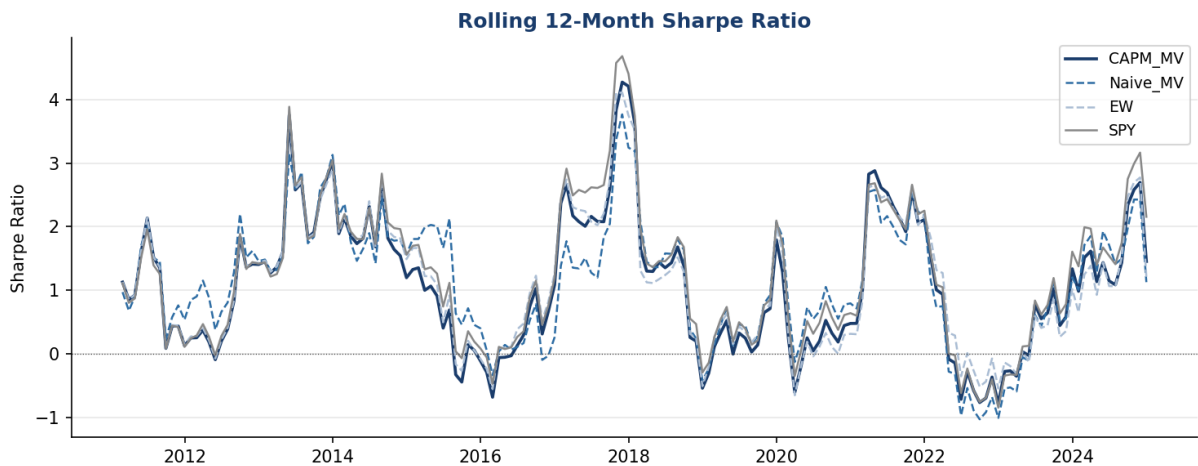


Figure 5: Rolling 12-month Sharpe ratio for all four strategies.

5.5 Historical VaR and Expected Shortfall

Table 5 and Figure 6 report one-month historical 95% VaR and ES for each strategy, using the positive-loss convention. The values are deliberately interpreted as empirical diagnostics rather than parametric tail forecasts: with 178 monthly observations, the 95% ES estimate is driven by roughly the worst nine months.

Table 5: Historical one-month 95% VaR and Expected Shortfall. Values are positive loss percentages.

Strategy	1M VaR 95%	1M ES 95%	Worst Month
CAPM-MV	7.17%	9.99%	−17.17%
Naive MV	7.12%	9.18%	−13.45%
EW	6.05%	9.51%	−16.83%
SPY	6.61%	9.19%	−13.92%

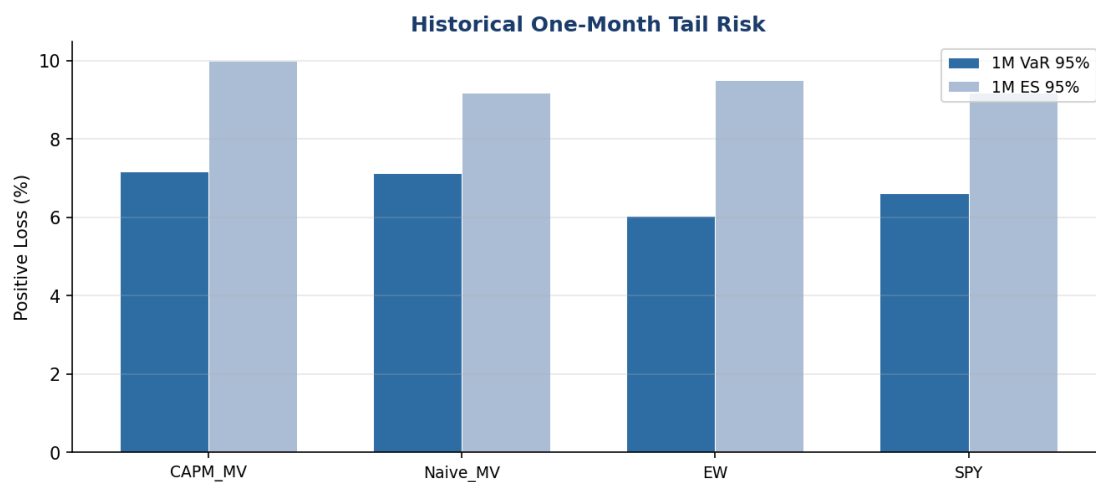


Figure 6: Historical one-month 95% VaR and Expected Shortfall for all four strategies. Values are shown as positive loss percentages.

CAPM-MV and Naive MV have similar one-month VaR at approximately 7.2%, but CAPM-MV has a higher ES of 9.99% compared with 9.18% for Naive MV. This does not contradict CAPM-MV’s superior maximum drawdown: maximum drawdown is path-dependent, whereas ES averages the worst monthly returns. The result shows that CAPM-MV reduced the depth of the 2022 drawdown relative to Naive MV, but its worst individual months, particularly around the COVID crash, were more severe.

6. Stress Testing and Sensitivity Analysis

6.1 Historical Stress Scenarios

Table 6 and Figure 7 report cumulative returns across three historical drawdown episodes.

The three scenarios tell distinct stories. During the COVID crash, CAPM-MV performed similarly to Equal Weight and worse than SPY: when all sectors fall simultaneously in a correlated sell-off, sector allocation matters less than aggregate market exposure, and the beta-structured tilt into cyclicals provided no protection. During the 2022 rate shock, the picture reverses sharply: CAPM-MV lost −19.4% versus −25.2% for Naive MV, a 5.8

Table 6: Cumulative returns (%) during stress scenarios. All figures net of costs.

Scenario	Period	CAPM-MV	Naive MV	EW	SPY
COVID Crash	Feb–Mar 2020	−26.4	−21.7	−26.1	−22.2
2022 Rate Shock	Jan–Dec 2022	−19.4	−25.2	−12.5	−20.2
2011 Debt Crisis	Aug–Oct 2011	−2.9	−0.6	−2.6	−2.5

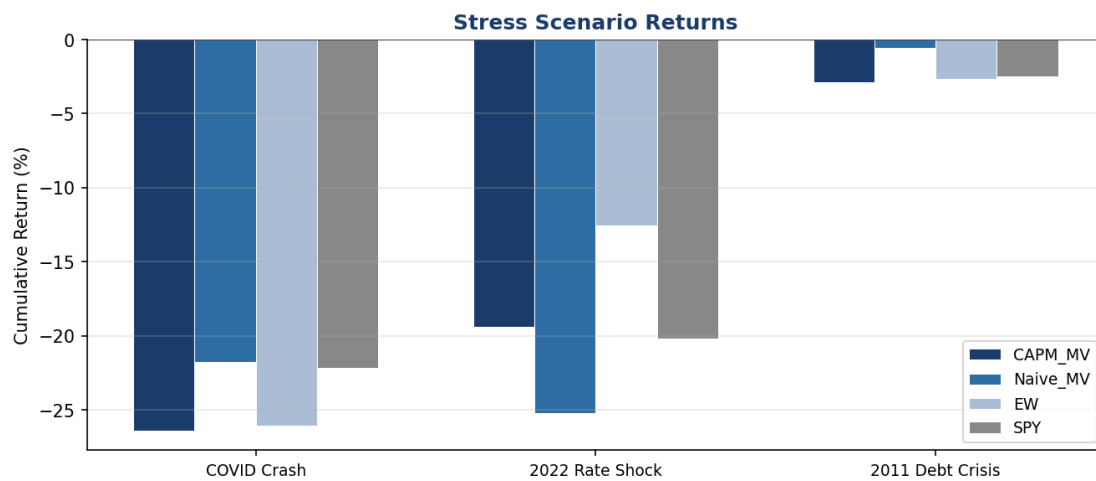


Figure 7: Cumulative returns (%) across the three stress scenarios.

percentage point advantage. The CAPM return structure, by anchoring expected returns to equilibrium betas rather than recent return momentum, maintained a more balanced sector allocation going into the rotation and avoided the deep exposure to growth sectors that damaged Naive MV. Equal Weight performed best in 2022 (−12.5%) by avoiding any active tilt altogether. The 2011 debt crisis shows negligible differences between strategies, as it was a short-lived risk-off episode in which sector selection had minimal impact.

6.2 Parametric Sensitivity Analysis

Figure 8 shows Sharpe ratios as risk aversion λ and the lookback window are varied one-at-a-time around the base case.

Two findings stand out. First, Sharpe increases monotonically with λ : higher risk aversion forces a more diversified, lower-volatility portfolio which in this constrained sector universe outperforms more concentrated tilts. At $\lambda = 3.0$, Sharpe reaches 0.813, approaching the Naive MV figure of 0.819, while one-month VaR and ES fall to 5.88% and 8.78% respectively. This suggests that much of the full-period underperformance relative to Naive MV can be attributed to parameter choice rather than to a fundamental deficiency in the CAPM return structure.

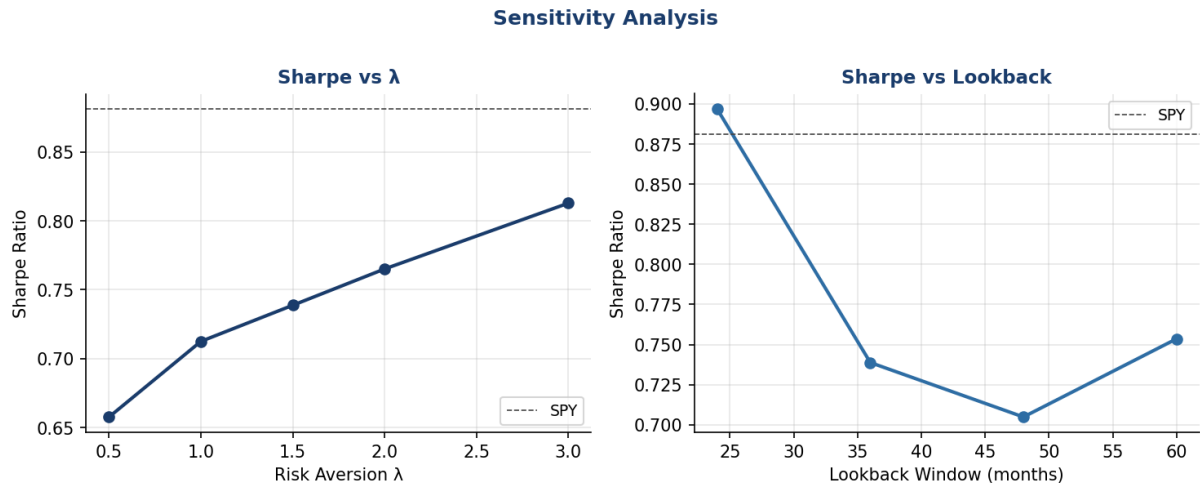


Figure 8: One-at-a-time sensitivity of Sharpe ratio to risk aversion λ (left) and estimation lookback window (right). Dashed line shows SPY Sharpe of 0.881.

Table 7: Sensitivity of Sharpe ratio, maximum drawdown and one-month tail risk to key parameters.

Parameter	Value	Sharpe	Max DD (%)	1M VaR 95%	1M ES 95%
λ sensitivity	0.5	0.658	−29.7	7.33%	10.78%
λ sensitivity	1.0	0.713	−26.8	7.38%	10.34%
λ sensitivity	1.5	0.739	−25.7	7.17%	9.99%
λ sensitivity	2.0	0.765	−26.0	6.90%	9.64%
λ sensitivity	3.0	0.813	−23.4	5.88%	8.78%
Lookback sensitivity	24m	0.897	−26.4	6.94%	9.53%
Lookback sensitivity	36m	0.739	−25.7	7.17%	9.99%
Lookback sensitivity	48m	0.705	−25.6	6.67%	9.83%
Lookback sensitivity	60m	0.754	−24.7	6.34%	9.71%

Second, the 24-month lookback window produces a Sharpe of 0.897, which exceeds SPY's 0.881 and is materially above the 36-month base case, with one-month VaR and ES of 6.94% and 9.53%. Shorter beta windows are more responsive to regime changes, and that responsiveness appears to be rewarded in this universe over this period. This is the most actionable finding in the sensitivity analysis and warrants systematic investigation as the primary extension to this work.

7. Discussion

7.1 Interpreting the CAPM-MV vs Naive MV comparison

The most important result in this study is the comparison of CAPM-MV against Naive MV, not against SPY. Over the full 2010–2024 period, CAPM-MV underperforms Naive MV on Sharpe (0.739 vs 0.819), but the full-period aggregate masks a regime-dependent pattern: Naive MV leads in 2010–2019 while CAPM-MV leads in 2020–2024. This is consistent with the theoretical motivation for using CAPM betas. In a steady trending market—the post-GFC recovery and the pre-COVID bull run—historical mean returns carry genuine signal and the Naive MV return vector is informative. In a volatile, regime-shifting environment, historical means become noisy and the structured CAPM prior reduces estimation error enough to add value.

The theoretical case for CAPM-MV over Naive MV rests on one argument: the sample mean of 36 monthly returns is a poor estimate of expected return for a single asset (standard error $\approx \sigma/\sqrt{36}$, which for a typical sector ETF with $\sigma \approx 18\%$ p.a. gives a monthly standard error of roughly 3%). The CAPM estimate replaces this with a model-implied value whose standard error is driven by uncertainty in $\hat{\beta}$ rather than in the mean return itself. Beta estimates from 36 months are far more precise than mean estimates from the same window.

What the CAPM structure does not provide is any view on which sectors will outperform the market. Expected returns are determined entirely by beta and the fixed ERP; there is no momentum signal, no valuation input, no stock selection. The strategy is closer in spirit to a risk-model-driven smart-beta allocation than to a return-forecasting approach. The results are consistent with this characterisation: CAPM-MV offers drawdown protection (Calmar 0.447 vs Naive MV 0.399) at the cost of some return in trending regimes.

7.2 Where the strategy underperforms and why

The primary mechanism behind CAPM-MV’s underperformance in 2010–2019 is the lag in rolling beta estimates. The post-GFC decade was dominated by the technology sector’s sustained outperformance. Historical mean returns for XLK were consistently high, causing Naive MV to overweight technology—a bet that paid off. CAPM-MV’s beta-structured return for XLK rose more slowly as 36-month betas updated gradually, resulting in a lower technology allocation that missed the sustained growth period.

The 2020–2024 reversal illustrates the same mechanism working in CAPM-MV’s favour: entering 2022, Naive MV’s historical mean returns were skewed toward growth sectors that had performed well in 2020–2021, producing a portfolio heavily exposed to the sectors that fell most sharply in the rate-shock rotation. CAPM-MV’s more stable beta-based allocation was less concentrated in these sectors.

7.3 Sector weight dynamics

The weight heatmap (Figure 3) confirms the optimiser is genuinely responding to beta variation rather than holding a near-equal-weight portfolio. The systematic decline in XLE, XLU, and XLRE weights and the rise in XLK and XLY weights across the 15-year period reflect real changes in the cross-sectional beta structure. These are economically meaningful tilts, not noise: each sector movement has a corresponding explanation in the

macro environment over that period.

8. Limitations and Extensions

8.1 Limitations

- **Single-factor CAPM.** The return model ignores well-documented risk factors beyond the market (value, momentum, quality, low-volatility). A Fama–French 3- or 5-factor model [6] would produce richer cross-sectional return predictions and potentially reduce the lag that caused underperformance in trending regimes.
- **Fixed ERP.** A constant 5.5% ERP ignores time-variation in the equity risk premium. A forward-looking implied ERP from cyclically-adjusted earnings yields would be more responsive to valuation regimes and introduce a genuine market-timing dimension to the return signal.
- **Normality assumption.** The mean–variance framework implicitly assumes quadratic utility or normally distributed returns. Sector ETFs exhibit skewness and fat tails, particularly in drawdown periods. CVaR optimisation would be a theoretically superior objective for an investor concerned with tail risk.
- **Limited universe.** The eleven sector ETFs are by construction highly correlated to the broad market and to each other. The diversification benefit is more constrained than it would be with factor ETFs, international equities, or multi-asset constituents. This limits how far the optimiser can deviate from SPY-like performance and partially explains why all active strategies underperform the passive benchmark over the full period.
- **No regime-detection mechanism.** The strategy has no explicit mechanism to reduce risk in adverse market conditions. A VIX-based risk-off overlay could materially improve stress-period performance, particularly during the correlated sell-off of the COVID crash.
- **XLC/XLRE backfill.** The SPY-scaled backfill for XLC (pre-2018) and XLRE (pre-2015) assigns these tickers a beta of 1.0 in early windows, understating their true beta dispersion in that period.

8.2 Possible Extensions

- **Shorter lookback window.** The sensitivity analysis finds that a 24-month lookback window produces a Sharpe ratio of 0.897, exceeding SPY’s 0.881. This is the most immediately actionable finding: shorter windows make beta estimates more responsive to regime changes, which appears to be rewarded in this universe. A systematic investigation of adaptive lookback selection is warranted.
- **Higher risk aversion.** The sensitivity analysis shows that $\lambda = 3.0$ produces a Sharpe of 0.813 with a maximum drawdown of -23.4% . A more conservative parameterisation improves both metrics simultaneously in this constrained universe, suggesting the base case $\lambda = 1.5$ is too aggressive.

- **Multifactor return model.** Replace CAPM betas with Fama–French 5-factor loadings to improve cross-sectional return predictions and reduce the omitted-factor bias that likely drove underperformance in the growth-dominated 2010–2019 period.
- **Black–Litterman integration.** Use the CAPM equilibrium return as the prior and layer in momentum or valuation signals as views [7], inheriting Bayesian shrinkage properties while introducing a forward-looking signal dimension absent from the current framework.
- **CVaR optimisation.** Replace the variance objective with Conditional Value-at-Risk, directly targeting tail risk rather than variance, which is more relevant for the fat-tailed return distributions observed in sector ETFs during stress periods.

9. Conclusion

This project implements and evaluates a CAPM-informed mean–variance portfolio optimisation strategy over the full set of US SPDR Sector ETFs in a strict walk-forward backtest using a January 2010 start convention and 178 realised monthly returns from March 2010 to December 2024. The central methodological choice—replacing historically estimated returns with CAPM-structured expected returns—is motivated not as a return-forecasting claim but as a regularisation: imposing cross-sectional structure on the return input to reduce the estimation error that drives the worst failures of Markowitz optimisation in practice.

Over the full period, CAPM-MV achieves a Sharpe ratio of 0.739, below SPY (0.881) and Naive MV (0.819), but with materially better drawdown protection than Naive MV (−25.7% vs −30.7%) and a higher Calmar ratio relative to Naive MV (0.447 vs 0.399). Its historical one-month 95% VaR and ES are 7.17% and 9.99%, respectively, showing that drawdown improvement does not translate into uniformly lower monthly tail losses. The full-period Sharpe comparison between CAPM-MV and Naive MV masks a regime-dependent pattern that is the study’s most important finding: CAPM-MV underperforms Naive MV in the steady trending markets of 2010–2019 but outperforms in the volatile 2020–2024 period. This is precisely consistent with the theoretical motivation—the CAPM prior adds value when historical means are most unreliable, and loses value when recent history is genuinely informative.

The sensitivity analysis identifies two actionable findings. First, a 24-month lookback window raises Sharpe to 0.897, exceeding SPY, by making beta estimates more responsive to regime changes. Second, higher risk aversion ($\lambda = 3.0$) raises Sharpe to 0.813 with lower drawdown (−23.4%) and lower historical ES (8.78%), suggesting the base parameterisation is too aggressive for this constrained universe. These findings define a clear direction for further development.

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